

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE-AI	Date:	20/08/2026

Code of Conduct

I G.Vinay Kumar Reddy(192472109) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: vinay

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether **"to the car"** is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates **"with AC sleeper"** with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations

- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, **"with two passengers"** may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- Analyze the possible syntactic structures and ambiguities in the given voice command.
- Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

		performance evaluation			
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1	3	3	3	3	4	3	16	78
2	3	3	3	3	4	3	16	
3	3	3	3	3	4	3	16	
4	3	3	3	3	4	3	16	
5	3	3	3	3	4	3	16	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. AI-Powered Insurance Claim Processing System

a) Structural Ambiguity using CFG

Sentence: “I reported the damage to the car after the accident.”

Two possible interpretations are:

1. “to the car” modifies “damage”
→ The damage was to the car.
2. “to the car” modifies “reported”
→ The phrase is treated as a destination/recipient of reporting.

Simplified CFG:

$S \rightarrow NP VP$

$VP \rightarrow V NP PP$

$NP \rightarrow Det N$

$PP \rightarrow P NP$

The ambiguity occurs because the PP “to the car” can attach to either the noun phrase or verb phrase.

b) Limitations of Basic CFG

- Cannot effectively resolve structural ambiguity.
- Does not assign probabilities to different parses.
- Cannot naturally handle subject–verb agreement.
- Performs poorly with long and nested sentences.
- Has difficulty with incomplete and conversational input.

c) Improved Parsing Approach

Customer Input



Preprocessing



Earley Parser



PCFG Ranking



Feature Structure Checking



Claim Information Extraction

- **PCFG:** selects the most probable parse.
- **Feature Structures:** check number, person, tense, and semantic roles.
- **Earley Parsing:** handles ambiguity, incomplete input, and complex sentences.

d) Benefits

The proposed system improves **accuracy** through probabilistic parsing, **reliability** through feature checking, and **scalability** through efficient Earley parsing. It can automatically extract claimant, damaged object, cause, and claim amount.

2. Intelligent Railway Ticket Booking Assistant

a) CFG Interpretation

Sentence:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

Possible structure:

VP

|— Book
|— two tickets
|— from Chennai to Delhi
|— for my parents
|— with AC sleeper

The intended interpretation is that **“with AC sleeper” describes the ticket class**, not the parents.

b) Problems with Top-Down Parsing

Top-down parsing may require excessive **backtracking** when multiple grammar rules are possible. It also struggles with:

- Ambiguous phrase attachment.
- Incomplete requests.
- Long-distance dependencies.
- Long and complex queries.
- Real-time conversational input.

c) Earley Parsing

Earley Parsing maintains possible interpretations using **prediction, scanning, and completion**. It can process incomplete input and retain multiple parse structures until enough information is available to select the correct one.

PCFG can rank the alternatives and select:

with AC sleeper → Ticket Class

d) Comparison

Feature	Top-Down	CFG	Earley
Ambiguity	Poor	Moderate	Good
Backtracking	High	Moderate	Low
Incomplete input	Poor	Limited	Good
Complex queries	Limited	Moderate	Good
Real-time use	Less suitable	Moderate	Suitable

Conclusion: Earley Parsing is more suitable for conversational railway booking.

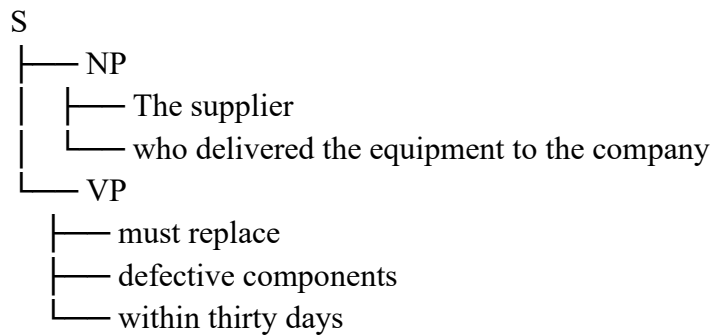
3. AI-Based Legal Contract Analysis System

a) CFG Structure

Sentence:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

Structure:



The main ambiguity is the attachment of **“to the company.”** The correct interpretation is that the **supplier delivered the equipment to the company.**

b) Limitations of Basic CFG

Basic CFG cannot reliably identify:

- Agent and recipient.
- Relative-clause relationships.
- Nested structures.
- Legal obligations.
- Conditions and deadlines.

c) Improved Architecture

Contract



Preprocessing



Earley Parsing



PCFG Ranking



Feature Structures



Semantic Role Extraction



Legal Information

d) Information Extraction

Information Extracted Value

Supplier	Supplier
Action	Replace
Equipment	Equipment
Company	Company
Obligation	Replace defective components
Deadline	Within 30 days

Benefit: The combination of CFG, PCFG, Feature Structures, and Earley Parsing improves the accuracy of legal entity, obligation, and deadline extraction.

4. AI-Based E-Commerce Product Search System

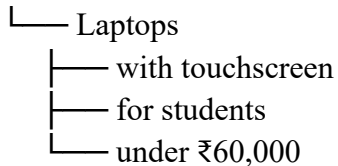
a) CFG Ambiguity

Query:

“Find laptops with a touchscreen for students under ₹60,000.”

Intended structure:

Find



The phrase **“for students”** may incorrectly be attached to **“touchscreen”** instead of the laptops.

b) CFG Limitations

CFG has difficulty with:

- Informal queries.
- Missing words.
- Ambiguous PP attachment.
- Long product descriptions.
- Elliptical queries such as:

“Laptop touchscreen under 60k.”

c) Improved Architecture

User Query



Preprocessing



Earley Parser



PCFG Ranking



Feature Structures



Product Attribute Extraction



Search Engine

The system extracts:

Product = Laptop

Feature = Touchscreen

Audience = Students

Price = $\leq ₹60,000$

d) Benefits

- **PCFG** improves ambiguity resolution.
- **Feature Structures** identify product attributes.
- **Earley Parsing** handles incomplete queries.
- Structured attributes improve search precision and response time.

5. Intelligent Airline Voice Assistant

a) Syntactic Ambiguity

Command:

“Change my flight to Mumbai tomorrow with two passengers.”

Correct interpretation:

Action → Change Flight

Destination → Mumbai

Date → Tomorrow

Passengers → 2

The phrase **“with two passengers”** may incorrectly be attached to Mumbai instead of the booking information.

For:

“Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

the system must replace **Mumbai** with **Bangalore**.

b) Top-Down Parsing Limitations

Top-down parsing struggles with:

- Excessive backtracking.
- Incomplete commands.
- Ambiguous phrases.
- Speech recognition errors.
- User corrections.
- Real-time response requirements.

c) Improved Architecture

Voice Input



Speech Recognition



Earley Parser



PCFG Ranking



Feature Structures



Dialogue Management



Booking Request

- **Earley:** handles partial and ambiguous speech.
- **PCFG:** selects the most likely interpretation.
- **Feature Structures:** maintain destination, date, passenger count, etc.
- **Dialogue Manager:** handles corrections.

d) Processing Workflow

Voice



Text



Parsing



Ambiguity Resolution



Feature Extraction



Dialogue Correction



Booking System

For the correction:

Mumbai → Bangalore

Passengers → 2

Date → Tomorrow

Overall Conclusion

CFG provides the basic grammatical structure, **PCFG** resolves ambiguity using probabilities, **Feature Structures** enforce grammatical and semantic constraints, and **Earley Parsing** handles complex and incomplete input efficiently. Therefore, the combined approach is more accurate, reliable, and suitable for real-time NLP applications.